

D.3 Mapping Object Class Labels to Tags

Matterport3D includes a raw class label for all labeled objects. For example the raw labels “arm chair” and “bean bag chair” are grouped into the common object class of “chair”. Both “arm chair” and “bean bag chair” are within the set of classes able to be recognized by the tagging model. Because of this, for each common object class, we identify the corresponding raw classes and map them to matching tags within the vocabulary of the tagging model. We also manually identified tags that corresponded to a common object class and included them in the mapping. The mapping for the common object classes to tags are defined as follows:

bathhtub: bath, jacuzzi

bed: bed, bed frame, bunk bed, canopy bed, cat bed, dog bed, futon, hammock, headboard, hospital bed, infant bed, mattress

cabinet: armoire, bathroom cabinet, cabinet, cabinetry, closet, file cabinet, kitchen cabinet, medicine cabinet, side cabinet, tv cabinet, wine cabinet

chair: armchair, beach chair, bean bag chair, beanbag, chair, computer chair, feeding chair, folding chair, office chair, rocking chair, swivel chair, throne

chest_of_drawers: bureau, drawer, dresser, nightstand

clothes: baby clothe, baseball hat, bathrobe, bathroom accessory, bikini, bikini top, blouse, christmas hat, cloak, clothing, coat, cocktail dress, corset, costume, cowboy hat, crop top, denim jacket, dress, dress hat, dress shirt, dress shoe, dress suit, evening dress, fur coat, gown, halter top, hat, headdress, headscarf, hoodie, jacket, jeans, jockey cap, kilt, kimono, lab coat, lace dress, laundry, leather jacket, maxi dress, miniskirt, overcoat, pants, pantyhose, polo neck, polo shirt, raincoat, robe, safety vest, scarf, shirt, ski jacket, sports coat, straw hat, sun hat, suspenders, sweat pant, sweater, sweatshirt, t shirt, t-shirt, trench coat, underclothes, vest, visor, waterproof jacket, wedding dress, wrap dress

counter: bar, buffet, counter, counter top, island, kitchen counter, kitchen island, wet bar

cushion: pillow, throw pillow

fireplace: fireplace, mantle

gym.equipment: barbell, dumbbell, stationary bicycle, training bench, treadmill, weight

picture: art, art print, couple photo, decorative picture, drawing, family photo, group photo, movie poster, oil painting, photo, photo frame, picture, picture frame, portrait, poster, publicity portrait, reflection, wedding photo

plant: bush, flower, grass, houseplant, plant, tree

seating: bench, church bench, park bench, seat, window seat

shower: shower, shower door, shower head

sink: basin, bathroom sink, sink

sofa: couch, loveseat

stool: bar stool, footrest, music stool, step stool, stool

table: altar, billiard table, changing table, cocktail table, computer desk, dinning table, foosball, glass table, kitchen table, office desk, picnic table, poker table, round table, side table, stand, table, vanity, workbench, writing desk

toilet: bidet, toilet bowl, toilet seat

towel: bath towel, beach towel, face towel, hand towel, paper towel, towel

tv_monitor: bulletin board, computer monitor, computer screen, display, monitor, television, white-board

D.4 Mapping Region Class Labels to Tags

We map region labels directly to corresponding tags in the vocabulary of the tagging model whenever possible. Some region labels are made up of multiple concepts such as “porch/terrace/deck/driveway”. In these cases, we try to map each concept to a tag if possible.

balcony: balcony

bar: bar

bathroom: bathroom

bedroom: bedroom

classroom: classroom

closet: closet

dining room: dining room

garage: garage

hallway: hallway

kitchen: kitchen

laundryroom/mudroom: laundry room

library: library

living room: living room

meetingroom/conferenceroom: meeting room

office: home office, office

porch/terrace/deck/driveway: deck, driveway, porch, terrace

rec/game: recreation room

spa/sauna: sauna

stairs: stairs, stairwell

tv: cinema, home theater, theater

utilityroom/toolroom: utility room

workout/gym/exercise: gym

We relabeled regions labeled “familyroom” and “lounge” as “living room” since these labels described analogous regions. Similarly, we also relabeled regions labeled “toilet” as “bathroom”.

E Label to Text Mappings for Embedding-Based Methods

To evaluate embedding-based methods, the class labels are mapped to strings which are then encoded into text embeddings for querying the map embeddings.

E.1 OpenScene and OpenMask3D

We followed the authors of OpenScene and OpenMask3D and applied the prompting method of “a {} in a scene” for object class labels. We modified the prompt to be grammatically correct for the following object classes:

chest_of_drawers: a chest of drawers in a scene

clothes: clothes in a scene

gym_equipment: gym equipment in a scene

tv_monitor: a television monitor in a scene

For region classes, we directly used the label as the string, except for the following classes where we modified the string to either be more grammatically correct or more accurately reflect the class semantics:

laundryroom/mudroom: laundry room or mudroom

meetingroom/conferenceroom: meeting room or conference room

porch/terrace/deck/driveway: porch or terrace or deck or driveway

rec/game: recreation or game room

spa/sauna: spa or sauna

tv: cinema or home theater or theater

utilityroom/toolroom: utility room or tool room

workout/gym/exercise: gym

E.2 CLIP Viewpoint Retrieval

For viewpoint retrieval using CLIP, we mapped each label to a string and used the ImageNet80 ensemble prompting method from Radford et al. [6] to create the text embedding.

For object classes, we used the label directly as the string, except for the following classes where we modified the string to be more grammatically correct or more accurately reflect the class semantics:

chest_of_drawers: chest of drawers

counter: countertop

gym_equipment: gym equipment

tv_monitor: television monitor

Similarly for region classes, we used the label directly as the string except for the following classes:

laundryroom/mudroom: laundry room or mudroom

meetingroom/conferenceroom: meeting room or conference room

porch/terrace/deck/driveway: porch or terrace or deck or driveway

rec/game: recreation or game room

spa/sauna: spa or sauna